PROBLEM7:

PROBLEM STATEMENT:

. Student Performance Trend Analysis: Compare assessment scores across multiple examinations, calculate improvements and declines, identify consistently weak subjects, and classify students according to defined criteria.

OBJECTIVE:

To develop a C program that compares student marks across multiple examinations, calculates improvements and declines, identifies consistently weak subjects, and classifies students based on their average marks.

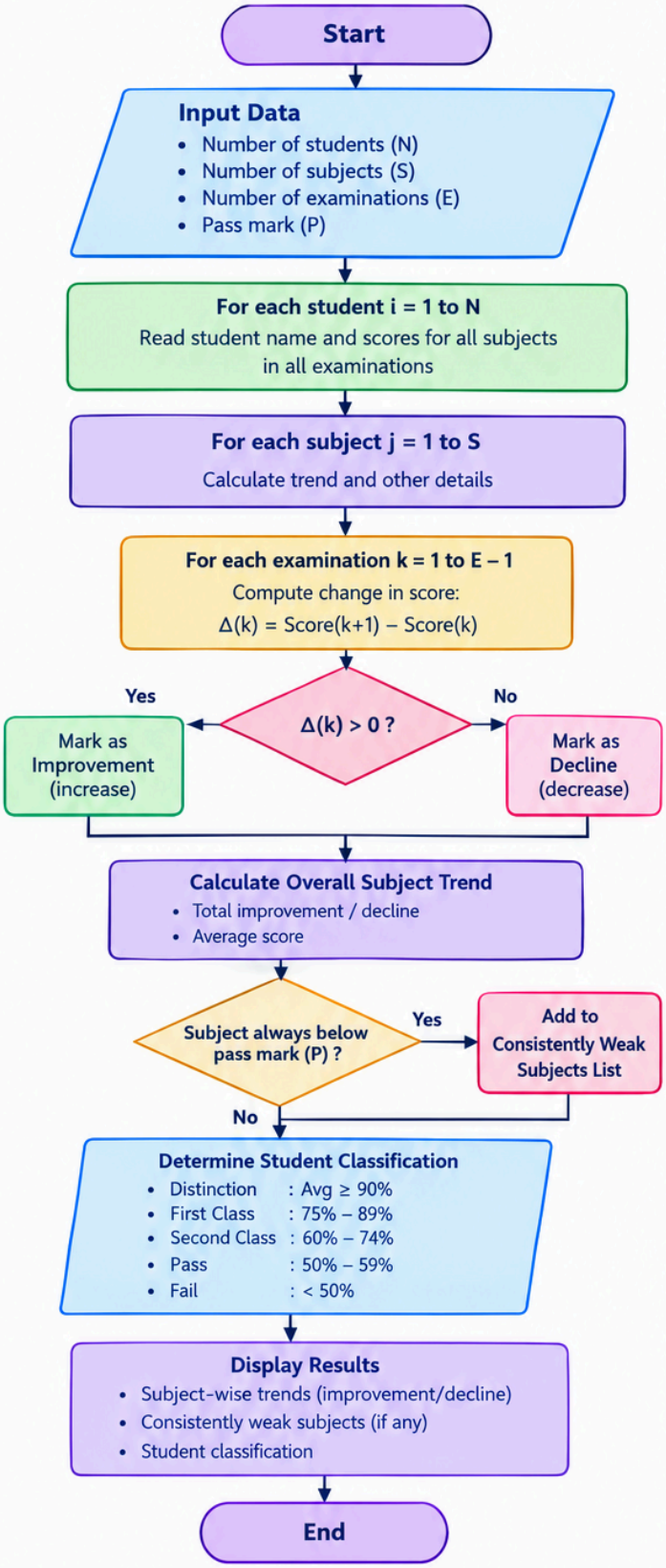
ALGORITHM:

1. Start the program.
2. Read the number of subjects and examinations.
3. Enter the subject names and marks obtained in each examination.
4. Validate the marks; each mark must be between 0 and 100.
5. Calculate the average marks for each examination.
6. Compare consecutive examination marks for every subject.
7. Display whether the marks improved, declined, or remained unchanged.
8. Identify subjects where marks are below 40 in every examination.
9. Calculate the overall average across all subjects and examinations.
10. Classify the student:
    - 90 and above — Excellent
    - 75 to below 90 — Very Good
    - 60 to below 75 — Good
    - 40 to below 60 — Average
    - Below 40 — Needs Improvement
11. Display the performance trends, weak subjects, overall average, and classification.
12. Stop.

FLOWCHART:

Student Performance Trend Analysis

(Compare assessment scores across multiple examinations, calculate improvements and declines, identify consistently weak subjects, and classify students.)



Example Input

Student	Subject	Ex1	Ex2	Ex3
Alice	Math	78	85	92
	Science	65	70	75
	English	58	62	68
Bob	Math	88	90	95
	Science	70	72	74
	English	45	48	52
Charlie	Math	50	55	60
	Science	40	42	45
	English	70	75	80

Pass mark (P) = 50

Key Rules

1. **Improvement / Decline**
   Δ = Score in next exam – Current score
2. **Consistently Weak Subject**
   Subject with all exam scores < Pass mark
3. **Student Classification**
   Based on average percentage of all subjects (or as per defined criteria).

Example Output (Summary)

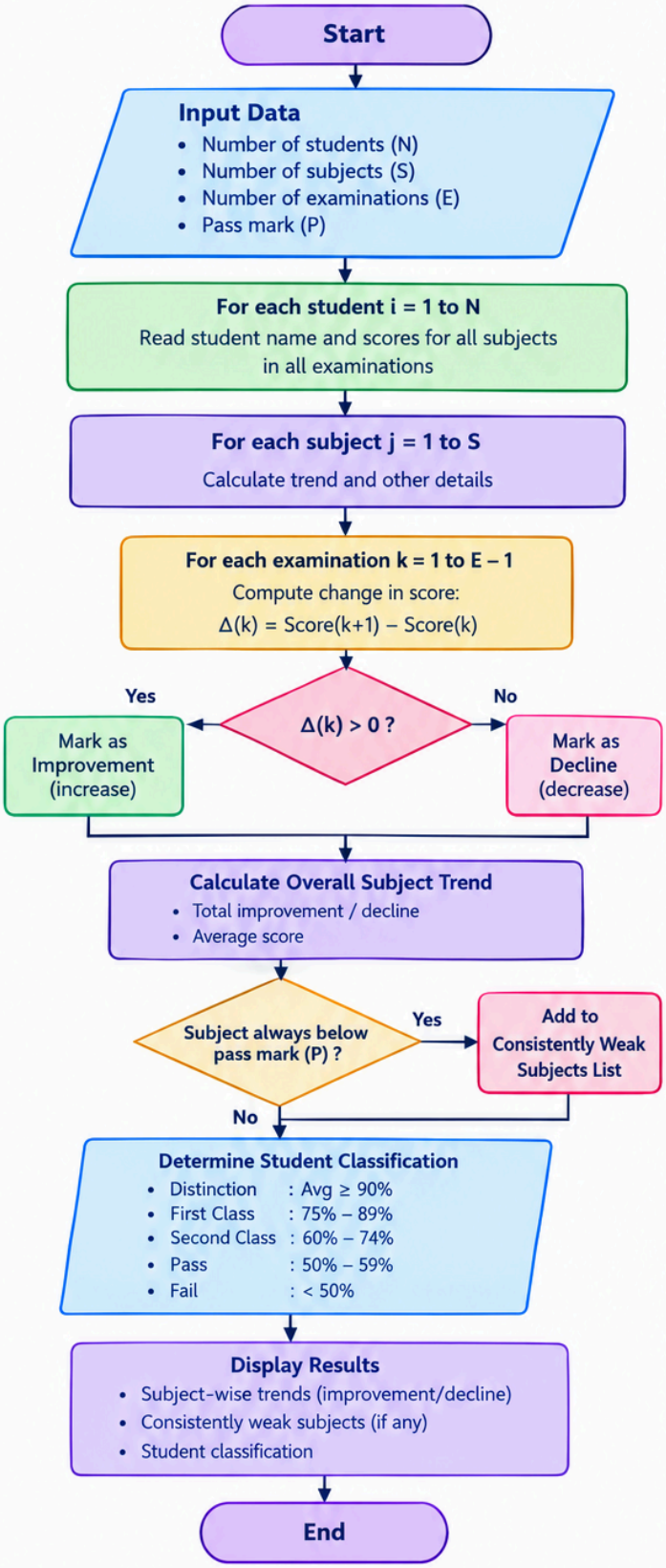
1. **Subject-wise Trend (Alice)**
   - Math : 78 → 85 → 92 (Improvement: +14)
   - Science : 65 → 70 → 75 (Improvement: +10)
   - English : 58 → 62 → 68 (Improvement: +10)
2. **Consistently Weak Subjects**
   - Charlie : None
   - Bob : English (45, 48, 52) → Weak
   - Alice : None
3. **Student Classification**
   - Alice : First Class (Avg: 78.3%)
   - Bob : First Class (Avg: 76.7%)
   - Charlie : Pass (Avg: 55.0%)

Note

- Improvement/decline is calculated between consecutive exams.
- A subject is weak if all exam scores are below the pass mark.
- Classification can be adjusted as per your institution's criteria.

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PROGRAM:

```
#include <stdio.h>

int main() {
    int n, e, i, j;
    char subjects[20][30];
    float marks[20][10], total = 0, average;
    float examTotal[10] = {0};

    printf("Enter number of subjects (max 20): ");
    scanf("%d", &n);

    printf("Enter number of examinations (2-10): ");
    scanf("%d", &e);

    if (n < 1 || n > 20 || e < 2 || e > 10) {
        printf("Invalid number of subjects or examinations!\n");
        return 0;
    }

    for (i = 0; i < n; i++) {
        printf("Enter subject %d name: ", i + 1);
        scanf("%29s", subjects[i]);
    }

    for (i = 0; i < n; i++) {
        printf("\nSubject: %s\n", subjects[i]);

        for (j = 0; j < e; j++) {
            printf("Enter marks for Exam %d: ", j + 1);
            scanf("%f", &marks[i][j]);

            if (marks[i][j] < 0 || marks[i][j] > 100) {
                printf("Invalid marks! Enter marks from 0 to 100.\n");
                return 0;
            }

            examTotal[j] += marks[i][j];
            total += marks[i][j];
        }
    }
```

```
printf("\n--- EXAM AVERAGES ---\n");
for (j = 0; j < e; j++) {
    printf("Exam %d Average: %.2f\n",
          j + 1, examTotal[j] / n);
}

printf("\n--- SUBJECT PERFORMANCE TRENDS ---\n");

for (i = 0; i < n; i++) {
    printf("\n%s:\n", subjects[i]);

    for (j = 1; j < e; j++) {
        float difference = marks[i][j] - marks[i][j - 1];

        if (difference > 0)
            printf("Exam %d to %d: Improved by %.2f\n",
                  j, j + 1, difference);
        else if (difference < 0)
            printf("Exam %d to %d: Declined by %.2f\n",
                  j, j + 1, -difference);
        else
            printf("Exam %d to %d: No change\n",
                  j, j + 1);
    }

    int weak = 1;
    for (j = 0; j < e; j++) {
        if (marks[i][j] >= 40)
            weak = 0;
    }

    if (weak)
        printf("Status: Consistently weak subject\n");
    else
        printf("Status: Not consistently weak\n");
}

average = total / (n * e);

printf("\n--- FINAL PERFORMANCE ---\n");
printf("Overall Average: %.2f\n", average);

if (average >= 90)
    printf("Classification: Excellent\n");
```

```
    else if (average >= 75)
        printf("Classification: Very Good\n");
    else if (average >= 60)
        printf("Classification: Good\n");
    else if (average >= 40)
        printf("Classification: Average\n");
    else
        printf("Classification: Needs Improvement\n");

    return 0;
}
```

PSEUDOCODE:

```
BEGIN

INPUT number_of_subjects
INPUT number_of_examinations

IF input values are invalid
    PRINT "Invalid input"
    STOP
END IF

FOR each subject
    INPUT subject_name
END FOR

FOR each subject
    FOR each examination
        INPUT marks

        IF marks < 0 OR marks > 100
            PRINT "Invalid marks"
            STOP
        END IF

        ADD marks to examination total
        ADD marks to overall total
    END FOR
END FOR

FOR each examination
```

```
    exam_average = examination_total / number_of_subjects
    DISPLAY exam_average
END FOR

FOR each subject
    FOR each consecutive examination pair
        difference = current_marks - previous_marks

        IF difference > 0
            DISPLAY "Improved"
        ELSE IF difference < 0
            DISPLAY "Declined"
        ELSE
            DISPLAY "No change"
        END IF
    END FOR

    IF marks in every examination < 40
        DISPLAY "Consistently weak subject"
    ELSE
        DISPLAY "Not consistently weak"
    END IF
END FOR

overall_average =
    overall_total / (number_of_subjects * number_of_examinations)

IF overall_average >= 90
    classification = "Excellent"
ELSE IF overall_average >= 75
    classification = "Very Good"
ELSE IF overall_average >= 60
    classification = "Good"
ELSE IF overall_average >= 40
    classification = "Average"
ELSE
    classification = "Needs Improvement"
END IF

DISPLAY overall_average
DISPLAY classification

END
```

OUTPUT:

Example with three subjects and three examinations

Enter number of subjects (max 20): 3
Enter number of examinations (2-10): 3
Enter subject 1 name: Maths
Enter subject 2 name: Physics
Enter subject 3 name: Chemistry

Subject: Maths
Enter marks for Exam 1: 60
Enter marks for Exam 2: 75
Enter marks for Exam 3: 85

Subject: Physics
Enter marks for Exam 1: 35
Enter marks for Exam 2: 38
Enter marks for Exam 3: 30

Subject: Chemistry
Enter marks for Exam 1: 70
Enter marks for Exam 2: 65
Enter marks for Exam 3: 80

--- EXAM AVERAGES ---
Exam 1 Average: 55.00
Exam 2 Average: 59.33
Exam 3 Average: 65.00

--- SUBJECT PERFORMANCE TRENDS ---

Maths:
Exam 1 to 2: Improved by 15.00
Exam 2 to 3: Improved by 10.00
Status: Not consistently weak

Physics:
Exam 1 to 2: Improved by 3.00
Exam 2 to 3: Declined by 8.00
Status: Consistently weak subject

Chemistry:
Exam 1 to 2: Declined by 5.00

Exam 2 to 3: Improved by 15.00
Status: Not consistently weak

--- FINAL PERFORMANCE ---
Overall Average: 60.89
Classification: Good